HTTPS Setup Guide

Namecheap Domain · Cloudflare DNS · Vultr VPS · Ubuntu 24.04 · Nginx

Goal: Secure a web app after purchasing a domain, route the domain to a Vultr VPS, install a trusted TLS certificate, and configure Cloudflare so both connections use HTTPS.

Who this guide is for

Team members who have not previously configured a custom domain or HTTPS. The steps assume the site already runs on a Vultr VPS using Ubuntu 24.04 and Nginx, and the domain was purchased through Namecheap.

What you will need

- Access to the Namecheap account and domain settings
- Access to the Cloudflare account and the domain zone
- SSH/sudo access to the Ubuntu VPS
- The VPS public IP address
- The actual domain name (used below as `example.com`)

Use your real domain anywhere this guide shows example.com. Do not copy example.com literally into production configuration.

How the pieces fit together

Namecheap is the registrar where the domain was purchased. Cloudflare manages authoritative DNS and can proxy web traffic. Nginx accepts web requests on the VPS and forwards them to the application (for example, Flask/Gunicorn). A TLS certificate lets the server prove its identity and encrypt HTTPS traffic.

Important: Changing nameservers to Cloudflare does not, by itself, install a certificate on the VPS. The recommended configuration here uses a publicly trusted Let's Encrypt certificate on Nginx and Cloudflare's **Full (strict)** encryption mode.

Before you begin: safety checklist

- Confirm the website currently loads over HTTP before making changes.
- Keep an active SSH session open while changing firewall settings.
- Back up Nginx configuration before running Certbot.
- Do not delete or replace existing Nginx proxy settings without understanding them.
- Make changes in order and verify each stage before proceeding.

1. Point Namecheap nameservers to Cloudflare

Cloudflare must be authoritative for the domain before its DNS records and proxy settings take effect.

1. Sign in to Cloudflare and choose **Add a site** if the domain has not been added yet.
2. Select the plan you intend to use and let Cloudflare scan/import existing DNS records.
3. Cloudflare will provide two assigned nameservers, typically ending in `.ns.cloudflare.com`.
4. Sign in to Namecheap → Domain List → Manage the domain → Nameservers.
5. Choose **Custom DNS** and enter the exact two nameservers Cloudflare assigned. Save.
6. Return to Cloudflare and wait until the zone shows as active. DNS changes can take time to propagate.

Use the exact nameservers shown in your Cloudflare account. Do not copy nameservers from an example or another domain.

2. Create the Cloudflare DNS records

In Cloudflare, open the domain → **DNS** → **Records**. Ensure the domain and www hostname point to the VPS public IP.

Type	Name	Content	Proxy status
A	@	Vultr public IPv4	Proxied (orange cloud)

Type	Name	Content	Proxy status
A or CNAME	www	VPS IPv4 or root domain	Proxied (orange cloud)

Notes:

• The root domain is commonly represented by `@` in the DNS interface.
• A CNAME for `www` pointing to the root domain is also valid.
• If an AAAA record exists, it must point to a working IPv6 address configured on the VPS. Remove an incorrect or unused AAAA record; otherwise some visitors may fail to connect.
• Proxying can stay enabled for the steps in this guide. Let's Encrypt HTTP validation can work through Cloudflare when the DNS and origin connectivity are correct.

3. Check the VPS and Nginx

Connect to the Ubuntu 24.04 VPS over SSH. These checks are read-only and help confirm Nginx is healthy before making changes.

```
sudo systemctl status nginx --no-pager
sudo nginx -t
ls -l /etc/nginx/sites-enabled/
sudo grep -RniE 'server_name|listen 80|listen 443' /etc/nginx/sites-enabled/
```

Look for the server block that handles your site. It should include both hostnames, for example:

```
server_name example.com www.example.com;
```

> Do not replace your existing Nginx server block with a generic sample. It may contain important proxy_pass, WebSocket, header, upload-size, or application-specific settings.

4. Back up the Nginx configuration

Create a dated backup before allowing Certbot to edit Nginx files:

```
sudo tar -czf "$HOME/nginx-backup-$(date +%F-%H%M%S).tar.gz" /etc/nginx
```

Keep the backup in a safe location. It contains configuration details; treat it as internal infrastructure information.

5. Allow HTTP and HTTPS through the firewall

Let's Encrypt's standard HTTP validation needs inbound TCP port 80. Website HTTPS uses TCP port 443. Check UFW:

```
sudo ufw status verbose
```

If UFW is active and SSH access is already allowed, allow the Nginx Full profile:

```
sudo ufw allow 'Nginx Full'
```

Also check the Vultr Firewall (if attached to the VPS) and allow inbound TCP 80 and 443. Keep SSH allowed. If UFW is inactive, do not enable it blindly on a production server; review all existing rules and access requirements first.

Test HTTP from a computer outside the VPS:

```
curl -I http://example.com
curl -I http://www.example.com
```

A 200, 301, or 302 response may be normal. A timeout or Cloudflare 522 usually points to an origin connectivity issue that should be resolved before requesting the certificate.

6. Install Certbot

Certbot requests a Let's Encrypt certificate and can configure Nginx to use it. First check whether it is already installed:

```
certbot --version
snap list certbot
```

If Certbot is not installed, use the Snap package on Ubuntu 24.04:

```
sudo snap install --classic certbot
sudo ln -sf /snap/bin/certbot /usr/local/bin/certbot
sudo certbot --version
```

> If Certbot is already installed, check how it was installed before adding another installation method. Avoid mixing Snap and APT versions.

7. Request a certificate for both hostnames

Run Certbot with the Nginx plugin. Replace the example names with your real domain and www hostname:

```
sudo certbot --nginx \
  -d example.com \
  -d www.example.com
```

Follow the prompts to enter an email address, accept the terms, and complete validation. If Certbot offers to redirect HTTP traffic to HTTPS, choose the redirect option.

Expected result: Certbot reports that the certificate was successfully obtained and installed. If it fails, stop and save the full error output; do not keep retrying without fixing the cause.

Common reasons certificate issuance fails:

- The DNS records do not point to the correct VPS.
- The domain's nameserver change is not active yet.
- Port 80 is blocked by UFW, the Vultr Firewall, or another firewall.
- Nginx does not have a server block matching the requested hostname.
- An incorrect AAAA record sends validation traffic to a broken IPv6 endpoint.
- A different service is responding on port 80.

8. Test and reload Nginx

After Certbot succeeds, test the configuration and reload Nginx:

```
sudo nginx -t
sudo systemctl reload nginx
sudo certbot certificates
```

The certificate listing should include both example.com and www.example.com. It also shows the expiry date and certificate paths.

9. Set Cloudflare to Full (strict)

Only do this after the origin certificate has been installed successfully and the VPS is serving HTTPS.

1. In Cloudflare, open your domain → **SSL/TLS** → **Overview**.
2. Select **Full (strict)** as the encryption mode.
3. Open **SSL/TLS** → **Edge Certificates**.
4. Enable **Always Use HTTPS** if you want Cloudflare to redirect visitors who use HTTP.

> Do not use Flexible mode for this recommended setup. Full (strict) encrypts the Cloudflare-to-origin connection and validates the origin certificate. Switching to Full (strict) before the origin certificate is ready can cause Cloudflare error 526.

10. Verify HTTPS from outside the VPS

Run these commands from your computer or another machine outside the VPS:

```
curl -I https://example.com
curl -I https://www.example.com
```

Check the certificate presented for the root hostname:

```
echo | openssl s_client \
  -connect example.com:443 \
  -servername example.com 2>/dev/null \
  | openssl x509 -noout -subject -issuer -dates
```

Repeat with www.example.com if needed. Open both HTTPS URLs in a browser and confirm there are no certificate warnings and the application still works.

11. Test automatic renewal

Let's Encrypt certificates expire and must be renewed. Test the renewal process:

```
sudo certbot renew --dry-run
systemctl list-timers --all | grep -i certbot
```

A successful dry run confirms the renewal process can complete in test mode. Confirm the Certbot timer or other renewal mechanism is active. Do not assume renewal works just because the first certificate was issued successfully.

12. Troubleshooting quick reference

Symptom	Likely checks
Cloudflare 522 / timeout	VPS is reachable; ports 80/443 allowed in both UFW and Vultr Firewall; Nginx is running; DNS points to the correct IP.
Cloudflare 526	Full (strict) cannot validate the origin certificate. Check that Certbot installed a valid certificate covering the hostname and that Nginx serves it on port 443.
Too many redirects	Check Cloudflare SSL/TLS mode, Nginx redirect rules, and application proxy headers. Use Full (strict), not Flexible, for this guide.
Certbot validation fails	Check authoritative DNS, port 80 reachability, matching Nginx server_name, and any incorrect AAAA record. Review Certbot's complete error output.
Root works but www fails	Check that www has a valid DNS record and is included in the certificate and Nginx server_name.
Browser certificate warning	Inspect the certificate's hostname, issuer, expiry, and the Cloudflare proxy status. Check that you are visiting the intended hostname.

13. Final go-live checklist

- ■ Namecheap uses the two nameservers assigned by Cloudflare.
- ■ Cloudflare DNS root and www records point to the correct Vultr VPS.
- ■ Any AAAA record points to a working IPv6 endpoint or has been removed if incorrect.
- ■ Nginx configuration passes sudo nginx -t.
- ■ TCP 80 and 443 are reachable through the applicable firewalls.
- ■ Certbot certificate includes both the root domain and www hostname.
- ■ Both HTTPS URLs load without certificate warnings.
- ■ HTTP requests redirect to HTTPS.
- ■ Cloudflare SSL/TLS mode is Full (strict).
- ■ Certbot renewal dry run succeeds and renewal is scheduled.

Operational tip: Keep a record of the domain, DNS records, Nginx site configuration file, certificate renewal status, and who owns access to Namecheap, Cloudflare, and Vultr. Never share private keys, passwords, API tokens, or unredacted secrets in troubleshooting messages.

Official references

- Certbot instructions: https://certbot.eff.org/
- Cloudflare SSL/TLS modes: https://developers.cloudflare.com/ssl/origin-configuration/ssl-modes/
- Cloudflare DNS records: https://developers.cloudflare.com/dns/manage-dns-records/
- Ubuntu Server documentation: https://documentation.ubuntu.com/server/